VS-04 Estimation, Analysis, and Statistical Modeling

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VS-04 Estimation, Analysis, and Statistical Modeling

1. Purpose and value created

VS-04 turns clean microdata into statistics. Up to this point the chain has improved the data itself. Here the focus shifts to producing numbers that describe a population: estimates, weighted aggregates, indices, time series, and model results. VS-04 does not change microdata values; it computes statistics from them. The output is a new kind of object, and each result carries a measure of how accurate it is. The value created is **analytical insight and statistical credibility**: data becomes information about a population that policymakers, researchers, and society can trust. Every decision in this value stream balances methodological detail with transparency and reproducibility.

VS-04 produces validated statistics with initial confidentiality protection. It does not compose products and does not publish them. Combining several analyses into integrated and composite statistical products belongs to VS-05 Statistical Product Composition. Final disclosure control for publication belongs to VS-06 Dissemination and Service Delivery.

> **i** Signature discipline: methodological transparency
>
> Every estimate carries the method, parameters, weights, and accuracy measure (sampling error, confidence interval, or model diagnostic) that produced it, and its lineage to its inputs. Any published figure can therefore be regenerated and defended. Imputation uncertainty from VS-03 is reflected in the variance. Fitness for purpose and coherence with related statistics are explicit acceptance criteria. Initial statistical disclosure control is applied, so the output can potentially be released.

VS-04 corresponds to the business domain **Estimation and balancing** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

PLACEHOLDER: ArchiMate view VS-04 (a) value stream and capabilities

2. Position and state transition

VS-04 is the fourth primary value stream. It receives Processed Data from VS-03 and applies statistical methods, estimation techniques, and analytical models to turn it into representative results. It is the stage where data becomes statistical information.

VS-04 takes data from State #3 to State #4, analytical version. Before VS-04, the data is edited, complete, and enriched microdata. After VS-04, it is a registered, immutable analytical version of statistics: estimates and aggregates calculated, accuracy and quality indicators produced, lineage documented, internal review completed, and initial disclosure control applied. The quality gate, the minimum metadata set, and the protection of State #4 are described in *State #4: Statistics* and are not repeated here. State #4 has two gate moments; VS-04 passes the analytical gate, and VS-05 later creates the approved version.

VS-04 receives Processed Data together with the metadata VS-03 attached: edit lineage, derived-variable definitions, and imputation flags. The imputation flags matter most. Estimation must reflect the extra uncertainty that imputed values introduce, for example through variance estimation that accounts for imputation. The edit and derivation lineage is carried through, so every estimate stays traceable to its observed and imputed inputs.

From	To	Gate	Event out
#3 Processed Data	#4 Statistics (analytical)	Analytical gate of State #4 (approved method, weights, accuracy, model diagnostics, soundness and coherence, initial disclosure control, reproducibility, metadata)	*Output Data Registered*

PLACEHOLDER: ArchiMate view VS-04 (c) state handover and events: Processed Data (#3) from VS-03, the estimation and analysis service at the boundary (weighting, estimation, aggregation, modelling, validation, disclosure control), Statistics (#4, analytical) shared with VS-05

3. Stages

The stages describe the value that VS-04 adds in every production cycle. Together they form the execution process type of VS-04. In a mature steady state, the methods have been designed and the engines built in advance (see attribute 4). Execution then focuses on computing weights, estimates, and models, and on capturing the methodology, parameters, and accuracy measures, so that every output is reproducible and defensible.

Stage	Name	Value added
VS-04.1	Confirm Estimation and Analytical Approach	The method fits the design and the purpose
VS-04.2	Apply Weighting and Calibration	Bias from sampling, non-response, and coverage is reduced
VS-04.3	Estimate and Aggregate Data	Statistics and indicators are available, each with its accuracy
VS-04.4	Perform Model-Based Analysis	Model-based results complement direct estimates

Stage	Name	Value added
VS-04.5	Validate and Document Results	Results are sound and documented (State #4)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-04.1 Confirm Estimation and Analytical Approach

Value created: *The right, pre-approved method selected for this cycle's data and objective.*

Execution starts when the *Processed Data Registered* event arrives from VS-03. In a mature steady state the methodology has already been designed, so each cycle confirms or selects the pre-approved approach for the data at hand. The team verifies that the calibration totals and reference sources are current, and confirms that the target parameters and domains of estimation still apply. A significant change, such as a new domain or a methodology revision, goes back to Design instead of being adapted during the cycle.

Activity	Key outputs	SAF process	Other reference
Confirm or select the pre-approved estimation and analytical approach for the cycle	Confirmed approach for the cycle	PRD-2.5 Design processing and analysis	
Verify that calibration totals and reference sources are current	Reference source confirmation	DLM-9 Manage reference and master data	CSDA Reference Data; 2.1.4 Statistical register management
Confirm target parameters, reference populations, and domains of estimation	Confirmed estimation scope	PRD-2.4 Design frame and sample	
Flag methodology changes for design review	Methodology change exception (if any)	PRD-2.5 Design processing and analysis	GAMSO Quality Management

VS-04.2 Apply Weighting and Calibration

Value created: *Bias from sampling, non-response, and coverage corrected, so results represent the population.*

The weighting engine applies the design weights and the non-response adjustment. The calibration engine then calibrates to the confirmed population totals, using the designed auxiliary variables and enforcing the weight bounds. Calibration convergence and the resulting weight distribution are checked against the designed constraints. The calibration totals, auxiliary variables, and convergence diagnostics are recorded, so the weighting can be reproduced.

Activity	Key outputs	SAF process	Other reference
Compute design weights and the non-response adjustment	Adjusted weights	PRD-5.6 Calculate weights	1.3.5 Estimating population parameters
Calibrate to the confirmed population totals with the designed auxiliary variables	Calibrated weights	PRD-5.6 Calculate weights	1.3.5 Estimating population parameters
Check convergence and weight distribution against the bounds	Weight diagnostics	PRD-5.6 Calculate weights	

Activity	Key outputs	SAF process	Other reference
Record calibration totals, auxiliary variables, and diagnostics as lineage	Weighting lineage	PRD-5.6 Calculate weights; OVA-4 Process Data Management	SAF-P09

VS-04.3 Estimate and Aggregate Data

Value created: *Representative statistics and aggregates, each carrying its accuracy.*

The estimation engine computes the target parameters with the calibrated weights. The aggregation pipeline computes the specified aggregates across the designed domains, with the agreed units and rounding conventions. The variance estimation component computes sampling errors and confidence intervals. These reflect both the survey design and the imputation uncertainty signaled by the VS-03 imputation flags. Every estimate is produced with its methodology, weights, and accuracy measure attached.

Activity	Key outputs	SAF process	Other reference
Compute target estimates with the calibrated weights	Point estimates	PRD-5.7 Calculate aggregates	1.3.5 Estimating population parameters
Compute aggregates across the designed domains (units, rounding)	Output aggregates	PRD-5.7 Calculate aggregates	
Compute sampling errors and confidence intervals (design and imputation uncertainty)	Accuracy measures	PRD-5.7 Calculate aggregates	
Attach methodology, weights, and accuracy to every estimate	Estimate metadata	PRD-5.7 Calculate aggregates; OVA-2 Metadata Management	SAF-P09

PLACEHOLDER: VS-04 estimation and aggregation architecture: imputation-flagged microdata (#3) and calibrated weights feeding point estimation and aggregation by domain; variance estimation reflecting both the survey design and the imputation flags; every output carrying its accuracy and methodology

VS-04.4 Perform Model-Based Analysis

Value created: *Deeper insight: relationships, trends, and small-domain estimates that direct counts cannot give.*

This stage runs the model-based analyses specified in Design: analytical and regression models, time-series and forecasting models, and small-area models. Their diagnostics are computed and checked against the designed bounds, and seasonal adjustment is applied where applicable. Results are interpreted in context, and assumptions and limitations are documented. As with direct estimates, every model output carries its specification, parameters, diagnostics, and uncertainty. Model-based results are flagged distinctly from direct estimates.

Activity	Key outputs	SAF process	Other reference
Run the designed analytical, forecasting, and small-area models	Model outputs	PRD-6.1 Prepare draft outputs	1.4.1 Statistical (output) analysis

Activity	Key outputs	SAF process	Other reference
Compute and check model diagnostics against the designed bounds	Model diagnostics	PRD-6.2 Validate outputs	
Apply seasonal adjustment and time-series treatment where applicable	Adjusted series	PRD-6.1 Prepare draft outputs	
Interpret results; document assumptions and limitations	Analytical commentary	PRD-6.3 Interpret and explain outputs	1.4.1 Statistical (output) analysis

VS-04.5 Validate and Document Results

Value created: *Statistically sound, coherent, documented, and disclosure-safe statistics, ready for handover.*

Results are checked for statistical soundness and consistency, and for coherence with prior periods and related statistics. Significant movements are explained, and revisions are handled according to the revision policy. Initial statistical disclosure control is applied according to the designed policy: the disclosure risk is assessed, and protection methods such as suppression, rounding, or perturbation are applied, so the output can potentially be released. The methodology and quality indicators are documented. A persistent, immutable output version is created with a unique identifier and full lineage, registered in line with SAF-P09, and access permissions are applied in line with SAF-P10. The stage ends by publishing the *Output Data Registered* event, which tells VS-05 that new validated statistics are available for composition. If validation fails, the results are re-estimated or revised after review.

Output: Statistics (State #4), analytical version, the official handover artefact to VS-05. Its structure and minimum metadata set are described in *State #4: Statistics*.

Activity	Key outputs	SAF process	Other reference
Validate soundness and coherence; explain movements; handle revisions	Validation report	PRD-6.2 Validate outputs; DLM-7 Revise data	1.4.1 Statistical (output) analysis
Apply initial statistical disclosure control according to the designed policy	Disclosure-controlled outputs	PRD-6.4 Apply disclosure control	SAF-P11; 1.3.7 Statistical disclosure
Document methodology and quality indicators	Methodology and quality documentation	OVA-2 Metadata Management	SAF-P09
Pass the analytical gate and create an immutable, identified output version with full lineage	Versioned output data	DLM-2 Promote data through a quality gate; DLM-5 Store and version data	CSDA Traceability
Register metadata, apply access permissions, and publish the *Output Data Registered* event to notify VS-05	Catalogue registration, event	DLM-3 Register and catalogue data; DLM-4 Classify and protect data	SAF-P07, SAF-P10

4. The four process types

The sub-processes below are specific to VS-04. Each refines one or more SAF business processes, listed as *Refines*, which keeps it traceable to *Business Processes*. Sub-processes marked *(standard)* work as described

in Standard activities in every value stream; only the VS-04 specifics are listed.

Design: provides metadata

Without well-designed methods, estimation becomes ad hoc and indefensible, and management has no baseline to measure accuracy and coherence against.

Design sub-process	Refines	Main outputs
Design Estimation Strategy and Accuracy Approach	PRD-2.4 Design frame and sample; PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Estimation strategy, estimation specification, accuracy measurement plan, coherence criteria
Design Weighting and Calibration Methodology	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Weighting specification, calibration specification, weight constraint policy, calibration total source register
Design Aggregation and Model Specifications	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; OVA-2 Metadata Management	Aggregation specification, model specification, output statistic metadata standard, time-series treatment specification
Design Validation and Disclosure-Control Policy	PRD-2.5 Design processing and analysis; OVA-1 Quality Management; OVA-2 Metadata Management	Validation policy, disclosure-control policy, output documentation standard, reproducibility design
Design Quality-Measurement and Versioning Framework	OVA-1 Quality Management; DLM-1 Plan and design data assets; DLM-4 Classify and protect data	Quality indicator set, versioning policy, lineage specification, access control design

Design Estimation Strategy and Accuracy Approach

Refines: PRD-2.4 Design frame and sample; PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

The starting point is choosing how the statistics will be estimated. This sub-process selects the estimation strategy that fits the source and the analytical objective: design-based estimation for sample surveys, register-based aggregation, or model-assisted and model-based estimation for small domains. It defines the target parameters, the reference populations, and the domains of estimation. It also defines the accuracy measurement approach: how sampling errors and confidence intervals are computed, including how the uncertainty introduced by VS-03 imputation is reflected in variance estimation. Coherence criteria with related and prior statistics complete the design.

Activity	Key outputs	SAF process	Other reference
Select the estimation strategy per source and objective (design-based, register-based, model-based)	Estimation strategy document	PRD-2.5 Design processing and analysis	1.1.2 Design statistics product
Define target parameters, reference populations, and domains of estimation	Estimation specification	PRD-2.4 Design frame and sample; PRD-2.5 Design processing and analysis	

Activity	Key outputs	SAF process	Other reference
Define the accuracy measurement approach (sampling error, confidence interval), reflecting imputation uncertainty	Accuracy measurement plan	PRD-2.5 Design processing and analysis	
Define coherence criteria with related and prior statistics	Coherence criteria	OVA-1 Quality Management	GAMSO Quality Management

Design Weighting and Calibration Methodology

Refines: PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

This sub-process designs how sampling bias, non-response, and coverage errors are corrected through weighting. The team specifies the design weights, the non-response adjustment model, and the calibration method, for example calibration to known population totals from registers. It sets the auxiliary variables and calibration totals to use, and the authoritative source and version of those totals. Bounds and acceptance constraints on the weights prevent extreme weights.

Activity	Key outputs	SAF process	Other reference
Specify design weights and the non-response adjustment	Weighting specification	PRD-2.5 Design processing and analysis	1.3.5 Estimating population parameters
Specify the calibration method, auxiliary variables, and calibration totals with their sources	Calibration specification	PRD-2.5 Design processing and analysis	
Define weight bounds and acceptance constraints	Weight constraint policy	PRD-2.5 Design processing and analysis	
Define the authoritative source and version of the calibration totals	Calibration total source register	DLM-9 Manage reference and master data	CSDA Reference Data; 2.1.4 Statistical register management

Design Aggregation and Model Specifications

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; OVA-2 Metadata Management.

This sub-process specifies what is computed. For aggregation, each output statistic is defined by its formula, its domains and breakdowns, and its units and rounding conventions. For analysis, model-based methods (regression, time-series and forecasting, small-area models) are specified with their inputs, their assumptions, and the diagnostics that must be produced and checked. Seasonal adjustment and time-series treatment are specified where applicable. Each output statistic receives its metadata and classification, in line with SAF-P09.

Activity	Key outputs	SAF process	Other reference
Specify output aggregates: formulas, domains and breakdowns, units, rounding	Aggregation specification	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis	

Activity	Key outputs	SAF process	Other reference
Specify analytical and forecasting models: inputs, assumptions, diagnostics	Model specification	PRD-2.5 Design processing and analysis	
Define metadata and classification for every output statistic	Output statistic metadata standard	OVA-2 Metadata Management	SAF-P09
Define seasonal adjustment and time-series treatment where applicable	Time-series treatment specification	PRD-2.5 Design processing and analysis	

Design Validation and Disclosure-Control Policy

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management; OVA-2 Metadata Management.

This sub-process designs how results are validated and protected before handover. Validation criteria cover statistical soundness (plausibility, consistency, model diagnostics within bounds), coherence with related and prior statistics, and the revision policy. The statistical disclosure control policy sets the disclosure-risk criteria for tables and any output microdata, and the protection methods: suppression, rounding, and perturbation. The aim is an output that can potentially be released, in line with SAF-P11. Reproducibility is designed in: an output must be regenerable from the processed input plus the recorded methodology and parameters.

Activity	Key outputs	SAF process	Other reference
Define validation criteria: soundness, coherence, model diagnostics, revision policy	Validation policy	OVA-1 Quality Management	GAMSO Quality Management
Define disclosure-risk criteria and protection methods	Disclosure-control policy	PRD-2.5 Design processing and analysis	SAF-P11; 1.3.7 Statistical disclosure
Define output metadata and methodology documentation requirements	Output documentation standard	OVA-2 Metadata Management	SAF-P09
Design reproducibility: regenerate outputs from inputs, methodology, and parameters	Reproducibility design	PRD-2.5 Design processing and analysis	SAF-P10

Design Quality-Measurement and Versioning Framework

Refines: OVA-1 Quality Management; DLM-1 Plan and design data assets; DLM-4 Classify and protect data.

This last design sub-process makes sure that accuracy is measured and outputs are traceable. It defines the quality indicators to produce: sampling error, coefficient of variation, response and coverage rates, and model fit. It sets the output versioning policy: how versions are identified, what triggers a revision, immutability, and retention. It defines the lineage requirements from each statistic back to its processed inputs and the method applied, and the role-based access control for outputs, in line with SAF-P10.

Activity	Key outputs	SAF process	Other reference
Define quality indicators (sampling error, coefficient of variation, response and coverage rates, model fit)	Quality indicator set	OVA-1 Quality Management	GAMSO Quality Management
Define the output versioning and revision policy	Versioning policy	DLM-1 Plan and design data assets	CSDA Traceability
Define lineage from each statistic to its processed inputs and method	Lineage specification	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control for outputs	Access control design	DLM-4 Classify and protect data	SAF-P10; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-04 design process decomposition: inputs (processed data model with imputation flags, methodology set, calibration totals from registers, framework standards), the five design sub-processes in order, and the design document each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

Implementation sub-process	Refines	Main outputs
Build Weighting and Calibration Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	Configured weighting and calibration engines, calibration total integration, versioned methodology repository
Build Estimation and Aggregation Pipelines	PRD-3.2 Reuse or build processing and analysis components; OVA-2 Metadata Management	Estimation pipeline, variance component, aggregation pipeline, output metadata binding
Build Modelling Environment, Validation, and Disclosure-Control Tooling	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-5 Store and version data	Modelling environment, validation tooling, disclosure-control tooling, versioning store, events
Test and Validate the Pipeline *(standard)*	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalise production systems	Test reports for weighting, calibration, estimation, variance, aggregation, modelling, validation, disclosure control, and versioning store; reproducibility test (an output regenerated identically from the processed input plus the recorded methodology and parameters); benchmark or back-series test against known or prior-period results; accuracy measures and model diagnostics checked against expectations

Build Weighting and Calibration Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data.

This sub-process turns the weighting and calibration designs into operational tools. The weighting engine computes design weights and non-response adjustments. The calibration engine calibrates to the designed totals with the specified auxiliary variables and enforces the weight bounds. The connections to the authoritative register and reference sources are configured with version pinning, so the totals used are recorded and reproducible. All weighting logic is version-controlled, in line with SAF-P10.

Activity	Key outputs	SAF process	Other reference
Build or configure the weighting engine (design weights, non-response adjustment)	Configured weighting engine	PRD-3.2 Reuse or build processing and analysis components	SAF-P01
Build or configure the calibration engine with auxiliary variables and bounds	Configured calibration engine	PRD-3.2 Reuse or build processing and analysis components	
Configure calibration total connections to versioned register sources	Calibration total integration	PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	CSDA Reference Data
Version-control all weighting and calibration logic	Versioned methodology repository	PRD-3.4 Configure workflows	SAF-P10

Build Estimation and Aggregation Pipelines

Refines: PRD-3.2 Reuse or build processing and analysis components; OVA-2 Metadata Management.

The designed estimation and aggregation specifications are implemented as automated services. The estimation engine computes the target parameters with the designed accuracy measures. The variance estimation component computes sampling errors and confidence intervals, accounting for the survey design and, through the VS-03 imputation flags, for imputation uncertainty. The aggregation pipeline computes the specified aggregates across the designed domains with the units and rounding conventions. Every output is produced with its methodology, parameters, weights, and accuracy measure attached.

Activity	Key outputs	SAF process	Other reference
Build the estimation engine for the designed target parameters	Estimation pipeline	PRD-3.2 Reuse or build processing and analysis components	1.1.3 Build statistics product
Build the variance estimation component (design and imputation uncertainty)	Variance and accuracy component	PRD-3.2 Reuse or build processing and analysis components	
Build the aggregation pipeline (domains, units, rounding)	Aggregation pipeline	PRD-3.2 Reuse or build processing and analysis components	
Attach methodology, weights, and accuracy measure to every output	Output metadata binding	OVA-2 Metadata Management	SAF-P09

Build Modelling Environment, Validation, and Disclosure-Control Tooling

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-5 Store and version data.

This sub-process implements model-based analysis, validation, and disclosure control. The modelling environment runs the specified analytical, forecasting, and small-area models with their diagnostics and, where applicable, seasonal adjustment. Validation tooling implements the designed soundness and coherence checks, including comparison with prior periods and related statistics. Disclosure-control tooling implements the designed risk checks and protection methods: suppression, rounding, and perturbation. The output versioning store keeps each accepted output version immutable, identified, reproducible from inputs plus methodology, and linked to its lineage. Event notifications alert VS-05 when an output version is registered.

Activity	Key outputs	SAF process	Other reference
Build or configure the modelling environment (models, diagnostics, seasonal adjustment)	Modelling environment	PRD-3.2 Reuse or build processing and analysis components	
Implement validation tooling (soundness, coherence, prior-period comparison)	Validation tooling	PRD-3.2 Reuse or build processing and analysis components	
Implement disclosure-control tooling (risk checks, suppression, rounding, perturbation)	Disclosure-control tooling	PRD-3.2 Reuse or build processing and analysis components	SAF-P11; 1.3.7 Statistical disclosure
Build the output versioning store (immutable, identified, reproducible)	Versioning store	DLM-5 Store and version data	CSDA Traceability
Configure the *Output Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows	SAF-P07

PLACEHOLDER: VS-04 implementation pipeline: the five design artefacts feeding the three build sub-processes, followed by technical testing, reproducibility and benchmark testing, and accuracy verification, resulting in a production-ready estimation pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-04 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when a new Processed Data version is registered. The stages then run in order; VS-04.4 runs only where the design specifies model-based results. Execution applies the designed methods and the built engines as they are; a new domain or a methodology revision goes back to Design. Each run produces a new analytical version; preliminary, revised, and final figures are separate versions.

Aspect	VS-04
Trigger	*Processed Data Registered*
Cadence	Per production cycle, following the release calendar; additional runs for revisions according to the revision policy
Performed by	ROL-15 Subject-matter statistician (survey statistician, analyst or modeler, subject-matter reviewer), ROL-16 Disclosure control specialist, ROL-03 Data steward

Aspect	VS-04
Input	Processed Data (#3) with imputation flags and edit lineage, confirmed calibration totals, production-ready pipeline
Output	Registered analytical version of Statistics (#4) with methodology, accuracy, and lineage
Control point	Weight diagnostics in VS-04.2; analytical gate of State #4 in VS-04.5
Exception path	Validation failed: re-estimate or revise after review; methodology change back to Design; an error in the microdata back to VS-03 as a new #3 version

Management: provides quality reports

Management sub-process	Refines	Main outputs
Monitor Process Performance *(standard)*	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard; throughput report against the release calendar; paradata on methods, parameter and software versions run, and number of manual interventions
Assess Output Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Accuracy report, coherence report, revision analysis, quality trend analysis
Manage Methodology and Output-Version Lifecycle	OVA-2 Metadata Management; OVA-5 Knowledge Management; DLM-5 Store and version data; DLM-8 Monitor data quality, lineage and use	Methodology version register, metadata quality report, version store audit, metadata lineage audit
Audit and Compliance Reporting *(standard)*	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Reproducibility audit showing that every published figure can be regenerated from the processed input plus the recorded methodology and parameters; disclosure-control report on the control applied and the risk outcomes (SAF-P11); methodology compliance report confirming that approved estimation and analytical methods were applied (4.1.5 Accountability); security compliance report including the integrity of the tamper-evident methodology record (SAF-P10)

Management sub-process	Refines	Main outputs
Continuous Improvement *(standard)*	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, for example refining the calibration specification to reduce variance, recalibrating or replacing models, improving the non-response adjustment, adopting better variance estimators, and tuning the release process to improve timeliness; trends watched: sampling errors, revisions, coherence, model performance

Assess Output Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the outputs themselves is evaluated against the criteria set in *Design Quality-Measurement and Versioning Framework*. Each output is assessed on several dimensions. Accuracy covers sampling error, coefficient of variation, and model fit. Coherence covers consistency with related and prior statistics. Timeliness is measured against the release calendar. Reliability is assessed through revision analysis that compares preliminary and final figures. Quality indicators are produced per statistic, per output, and per cycle. Trends are tracked to detect deteriorating accuracy, growing revisions, or models that need recalibration. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate accuracy (sampling error, coefficient of variation, model fit)	Accuracy report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess coherence with related and prior statistics	Coherence report	OVA-1 Quality Management	CSDA Data Quality
Conduct revision analysis (preliminary compared with final figures)	Revision analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across statistics and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management

Manage Methodology and Output-Version Lifecycle

Refines: OVA-2 Metadata Management; OVA-5 Knowledge Management; DLM-5 Store and version data; DLM-8 Monitor data quality, lineage and use.

The estimation methods and the output versions they produce must stay current, complete, and well governed. This sub-process governs the life cycle of estimation methods, weighting and calibration specifications, and models. It versions them, manages transitions, such as a methodology change with back-series implications, and retires obsolete methods. It verifies that every output carries its full methodology, weights, accuracy measures, and lineage, as a compliance check on SAF-P09. It governs the output versioning store: retention, immutability, and reconciliation of catalogue entries with stored versions. Lineage from the processed state through estimation to the output state is tracked.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of estimation methods, weighting specifications, and models	Methodology version register	OVA-5 Knowledge Management; OVA-2 Metadata Management	SAF-P03
Verify metadata completeness (methodology, weights, accuracy, lineage)	Metadata quality report	OVA-2 Metadata Management	SAF-P09
Govern the output versioning store: retention, immutability, reconciliation	Version store audit	DLM-5 Store and version data	CSDA Traceability
Track metadata lineage from processed to output state	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability

PLACEHOLDER: VS-04 management feedback loop: data and paradata from Execution flowing through the five management sub-processes, with accuracy and timeliness dashboards to the process owner, accuracy, coherence, and revision reports to the quality manager, reproducibility and disclosure-control reports to VS-07, and updated methods and models back to Design

Metadata per process type

Metadata is produced in every process type, not only in Design. In this value stream, metadata is what makes a published figure defensible. The table shows the data metadata each process type of VS-04 contributes; the paradata are listed in Standard activities in every value stream. In Implementation, VS-04 also records the outcomes of reproducibility tests, and in Execution the parameter values and run identifiers of every run.

Process type	Data metadata produced
Design	Estimation and weighting methodology, calibration totals and their sources, aggregation specifications, model specifications, validation and disclosure-control policy
Implementation	Engine configurations, weighting and estimation parameters, model parameters, variance estimation settings, disclosure-control parameters
Execution	Per estimate: methodology, weights, accuracy measures (sampling error, confidence interval), model diagnostics, reference population, record of the disclosure control applied
Management	Accuracy and coherence assessments, revision analyses, lineage assessments, trend analyses

PLACEHOLDER: VS-04 business object lifecycle: design artefacts (estimation strategy, weighting and calibration specification, aggregation and model specifications, validation and disclosure-control policy) govern implementation assets (weighting and estimation engines, modelling environment, variance and disclosure-control tooling, versioning store), which turn Processed Data (#3) into Statistics (#4), generating methodology, weights, accuracy, and output versions that management assesses and feeds back to design

5. Actors and roles

The table shows the actors and the reference roles involved in VS-04. Specialised role names used in an NSI map onto these reference roles (see *Role names in the value streams*). For example, the estimation methodologist in the original VS-04 description is ROL-07 Methodologist, the model designer is ROL-06 Statistical designer, and the survey statistician, the analyst or modeler, and the subject-matter reviewer are all forms of ROL-15 Subject-matter statistician.

Actor or role	ID	Process type	Responsibility in VS-04
International statistical organisation (methodology authority)	ACT-07	External	Sets estimation, weighting, and analytical methods the NSI conforms to, for example Eurostat guidance
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-04
Data owner	ROL-02	Management	Accountable for the analytical version of State #4: methodological soundness and accuracy
Data steward	ROL-03	Execution	Registers output datasets and versions, manages metadata entries, applies access controls
Data custodian	ROL-04	Execution	Operates the output versioning store, access control, and retention
Statistical designer (model designer)	ROL-06	Design	Specifies analytical and forecasting models, their inputs, assumptions, and diagnostics
Methodologist (estimation methodologist)	ROL-07	Design	Designs the estimation strategy, the weighting and calibration methodology, and the accuracy measurement approach; owns the estimation, weighting, and model methods
Data engineer	ROL-10	Implementation	Builds and configures weighting, estimation, aggregation, and modelling engines and the pipeline

Actor or role	ID	Process type	Responsibility in VS-04
Platform engineer	ROL-11	Implementation	Configures estimation services, calibration total sources, and access controls
Test manager	ROL-12	Implementation	Plans and carries out testing of the estimation pipeline: technical, reproducibility, and benchmark tests
Subject-matter statistician (survey statistician)	ROL-15	Execution	Applies the designed weighting and calibration and computes estimates with accuracy measures
Subject-matter statistician (analyst or modeler)	ROL-15	Execution	Carries out model-based analysis, regression, and forecasting, and documents diagnostics
Subject-matter statistician (subject-matter reviewer)	ROL-15	Execution	Reviews results for plausibility, coherence, and fitness for purpose, and signs off
Disclosure control specialist	ROL-16	Design, Execution	Designs the disclosure-control rules and disclosure-risk criteria for outputs, and applies initial disclosure control
Quality manager	ROL-18	Management	Monitors accuracy and coherence indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures reproducibility and disclosure-risk compliance, and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-04 relies on, and what role they play. NSIs can use it to map the value stream to their own organisation.

Capability group	Capabilities	Role in VS-04
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the estimation and weighting methodology and the analytical models, and build the estimation pipeline
1.3 Processing	1.3.5 Estimating population parameters, 1.3.7 Statistical disclosure	Compute weights, estimates, and aggregates; apply initial disclosure control
1.4 Analysis	1.4.1 Statistical (output) analysis	Validate, interpret, and explain the results

Capability group	Capabilities	Role in VS-04
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage data, methodology metadata, and quality; assess accuracy and coherence; use registers for calibration totals
3.2 Computerisation and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain weighting, estimation, aggregation, and modelling engines
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning	Apply statistical disclosure control and assess output disclosure risk
4.1 Governance	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward estimation methodology, plan improvements, report on reproducibility and compliance

Stages, processes, and functions

The table shows which SAF business processes realise each stage of VS-04 and which business functions they mainly use.

Stage	Realised by processes	Main functions
VS-04.1 Confirm Estimation and Analytical Approach	PRD-2.4 Design frame and sample; PRD-2.5 Design processing and analysis	FUN-3.1 Design and build statistical products; FUN-4.1 Manage statistical and analytical methodology
VS-04.2 Apply Weighting and Calibration	PRD-5.6 Calculate weights	FUN-3.4 Estimate and analyse
VS-04.3 Estimate and Aggregate Data	PRD-5.7 Calculate aggregates	FUN-3.4 Estimate and analyse
VS-04.4 Perform Model-Based Analysis	PRD-6.1 Prepare draft outputs; PRD-6.3 Interpret and explain outputs	FUN-3.4 Estimate and analyse
VS-04.5 Validate and Document Results	PRD-6.2 Validate outputs; PRD-6.4 Apply disclosure control; OVA-2 Metadata Management	FUN-3.4 Estimate and analyse; FUN-3.5 Protect statistical confidentiality; FUN-7.2 Manage metadata

PLACEHOLDER: ArchiMate view VS-04 (b) stages realised by processes and functions

Framework mappings

This reference table maps each process type of VS-04 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.1, PRD-2.4, PRD-2.5, DLM-1, DLM-4, DLM-9, OVA-1, OVA-2	Phase 2 Design (2.1, 2.4, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality, Reference Data	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-5, DLM-9, OVA-2	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Processing and Analysis (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-2.4, PRD-2.5, PRD-5.6, PRD-5.7, PRD-6.1 to PRD-6.4, DLM-2 to DLM-5, DLM-7, DLM-9, OVA-2, OVA-4	Phase 5 Process (5.6 Calculate weights, 5.7 Calculate aggregates), Phase 6 Analyse (6.1 to 6.4); 2.4 and 2.5 in their recurrent form	Production (execution)	Data Analysis (execution)	1.3.5 Estimating population parameters, 1.3.7 Statistical disclosure, 1.4.1 Statistical (output) analysis, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Management	OVA-1, OVA-2, OVA-4, OVA-5, DLM-5, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-04 most directly.

Principle	Statement	Application in VS-04
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Estimation, weighting, and modelling tooling, such as calibration and variance estimation libraries, is open source where possible, and the NSI contributes to the communities that maintain it. The NSI keeps collective control over critical methodological capabilities instead of depending on vendors.
SAF-P07	Implement loosely coupled processes and systems	Estimation is independent from upstream editing and downstream composition. Statistics (State #4) is the decoupling interface: downstream consumers use validated statistics without knowing the estimation internals.
SAF-P08	Data is shared property	Output statistics and output microdata, once validated, are available to all authorised consumers. Estimation is not duplicated across products when the same validated output can serve several purposes.
SAF-P09	No data without metadata and classification	Every estimate carries metadata: the methodology and parameters applied, the weights used, accuracy measures (sampling error, confidence interval), the reference population, and a classification for derived statistics. This is what makes results transparent and reproducible.
SAF-P10	Security by design	Access controls, integrity verification, and a tamper-evident record of methods and parameters are built into the estimation pipeline from the start, so that every published figure is attributable and reproducible.
SAF-P11	Privacy by design	Initial statistical disclosure control is applied to outputs, and the disclosure risk of estimates and any output microdata is assessed by design, so the result can potentially be released.

Principles that apply across the process types

SAF-P09 is the backbone of methodological transparency: Design defines the methodology and the required output metadata, Execution captures the method, parameters, weights, and accuracy measures for every estimate, and Management assesses output quality and coherence. SAF-P10 governs the tamper-evident record of methods and the reproducibility of outputs across the stages VS-04.1 to VS-04.5. SAF-P11 applies especially to VS-04.5, where the disclosure risk of the output is

assessed before handover. SAF-P07 governs both the intake of State #3 from VS-03 and the handover of State #4 to VS-05.

8. Interfaces, services, and events

Boundary interfaces

VS-04 has two boundary interfaces: intake from VS-03, and handover to VS-05.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-03 → VS-04 (intake)	Registered Processed Data version (#3)	Minimum metadata set of State #3, in particular edit lineage, derived-variable definitions, imputation flags and methods, quality improvement assessment, quality version identifier, classification	*Processed Data Registered*	VS-03 keeps custody of the processed version; VS-04 takes responsibility for weighting, estimation, modelling, and validation on receipt of the event
VS-04 → VS-05 (handover)	Registered analytical version of Statistics (#4)	Minimum metadata set of State #4: methodology and parameters, weights, accuracy measures, model diagnostics, record of disclosure control applied, coherence results, quality indicators, output version identifier, lineage to the processed source, classification	*Output Data Registered*	VS-04 keeps custody of the output version; VS-05 takes responsibility for composing integrated and composite products on receipt of the event

Business services

Service	Offered to	Realised in	Interfaces
Weighting and calibration service	Estimation teams, consumers of processed data	VS-04.2 Apply Weighting and Calibration	Weighting and calibration engines
Estimation and aggregation service	Estimation teams, analysts	VS-04.3 Estimate and Aggregate Data	Estimation interface that returns estimates with accuracy measures; query interface for methodology and parameters

Service	Offered to	Realised in	Interfaces
Modelling and analysis service	Analysts, methodologists	VS-04.4 Perform Model-Based Analysis	Modelling environment
Validation and disclosure-control service	Quality managers, disclosure control specialists	VS-04.5 Validate and Document Results	Validation and disclosure-control tooling
Output data provisioning service	VS-05 Statistical Product Composition, data analysts, other statistical processes	VS-04.5 Validate and Document Results	Event notification (*Output Data Registered*) and a catalogue query interface to find and access output versions, their accuracy measures, and methodology
Output version catalogue service	All data consumers in the NSI	Management sub-process *Manage Methodology and Output-Version Lifecycle*	Search and browse interface for output versions, accuracy measures, methodology documentation, and lineage
Design and methodology service	Methodologists, statistical designers	All Design sub-processes of VS-04	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artefacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Processed Data Registered	Start of VS-04.1 for a new cycle
Output version accepted / *Validation failed*	Outcome of VS-04.5; failure starts re-estimation or revision after review
Output Data Registered	Completion of VS-04.5; triggers VS-05
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-04 event choreography between VS-03, NSI teams, and process management across Design, Implementation, Execution (Processed Data Registered, weighting, estimation, modelling, validation passed or failed, Output Data Registered), and Management (accuracy, timeliness, coherence, revisions, audit, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-04	Receives from VS-04
VS-07 Governance and Compliance	Methodology, privacy, and security policy	Reproducibility audits, disclosure-control and methodology compliance reports
VS-08 Innovation and Methodological Development	New estimation, weighting, and modelling methods	Improvement needs, for example better variance estimators or replacement models
VS-09 Infrastructure and Platform Enablement	Estimation and modelling environments, versioning store, catalogue, event interfaces	Capacity and environment requirements

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
On-time output rate	Share of outputs produced within the release calendar window	Per output, per cycle
Average sampling error or coefficient of variation	Typical accuracy of key estimates	Per statistic, per cycle
Coherence check pass rate	Share of outputs that pass the coherence checks on the first run	Per cycle
Revision magnitude	Size of revisions compared with previously published figures	Per statistic, per cycle
Reproducibility rate	Share of outputs that regenerate identically from inputs plus methodology	Per cycle
Processing time	Time from processed intake to versioned registration	Per output, per cycle

10. Examples

Labour Force Survey. The design-based estimation strategy for the employment rate calibrates to population totals by age, sex, and region, with a variance estimator (for example a linearisation or replication method) that accounts for the survey design and for the income and non-response imputation flagged in VS-03. Domains of estimation are national and NUTS-2. For cycle Q3-2026 the team confirms the strategy, the calibration totals from the latest population register extract, and the domains; no methodology change is needed, so the cycle does not return to Design. The employment rate is estimated at 71.4%, with a 95% confidence interval of ±0.5 percentage points at national level and wider intervals at NUTS-2. Each cell carries its estimate, coefficient of variation, and the method that produced it.

Household survey. The calibration is designed as a generalised regression (GREG) calibration to population totals by age, sex, and region from the population register, with weight bounds that cap the calibration factor. The engine pulls the totals from a pinned version of the register and records the totals and convergence diagnostics with every run. In production the calibration converges in 6 iterations, the weighted margins reproduce the register totals, and the calibration factor stays within the bounds.

Tax register to business statistics. For small NACE by region domains where the direct register aggregate is unstable, a small-area model borrows strength across domains to produce more reliable estimates. The model diagnostics (fit, prediction intervals) are recorded, and the model-based cells are flagged distinctly from direct aggregates.

Toolkit: SAF-TK-V04 VS-04 maturity checklist and scenarios